

Executive Summary

Overall Readmission Challenge

The hospital recorded 120,000 admissions and 112,587 discharges alive, with 14,210 30-day readmissions and an overall 30-day readmission rate of 12.62%. Only 1,006 readmissions occurred within 7 days, indicating that a substantial portion of the readmission burden occurs after the first week and through day 30. The rate was relatively stable across the available years and showed no substantial month-to-month variation, suggesting a persistent readmission challenge rather than a seasonal one. Following data-quality assessment and preparation, the datasets were considered suitable for the proposed analysis.

Where the Burden and Risk Are Concentrated?

The analysis shows that readmission risk and readmission burden are distinct. Several populations experienced substantially higher-than-average readmission rates, including patients associated with longer stays, sepsis, respiratory failure, previous admissions, and ICU care. At the same time, a large share of the hospital's total readmissions was concentrated within high-volume populations.

Secondary diagnoses accounted for 11,584 readmissions, or 81.52% of all 30-day readmissions, with a 15.97% readmission rate. Other major contributors included patients with low hemoglobin and emergency admissions. These findings indicate that readmission reduction should address both high-risk populations and high-burden populations, rather than relying on readmission rate alone.

What the Analysis Revealed?

Because secondary diagnoses represented the largest identified burden, a focused analysis examined additional factors associated with readmission within this population. Statistically significant differences were identified across clinical, utilization, admission, and discharge-related characteristics, with effect sizes used to assess the magnitude of the observed differences.

These findings can help clinicians and care-management teams identify populations associated with elevated readmission risk and help operational teams identify potential intervention opportunities. The results describe associations rather than causal relationships.

Prioritizing Patients for Intervention

Predictive modeling was then used to address a different operational question: which discharged patients should be prioritized when intervention resources are limited?

Logistic Regression provided the strongest combination of discrimination and interpretability, with an AUC of approximately 0.75. Adding 34 individual ICD-10 indicators improved AUC by only 0.0008, so the simpler baseline model was selected.

Percentile-based risk stratification showed that the highest-risk 10% of eligible patients accounted for 32.2% of observed 30-day readmissions in the test population, while the highest-risk 20% accounted for 50.7%. This demonstrates the potential to concentrate limited intervention resources on patients with substantially higher observed risk.

Recommendation

The findings support a targeted readmission-reduction strategy that combines population-level burden analysis, identification of associated clinical and operational factors, and patient-level risk stratification. Intervention resources can be prioritized toward higher-risk patients while high-burden populations and actionable admission and discharge pathways are addressed. The next step is to pilot targeted transitional-care interventions and evaluate whether they produce measurable reductions in subsequent readmissions.

The analysis answers three increasingly specific questions: Where is the problem? What factors are associated with it? And who should we prioritize for intervention? The final question—what intervention works and whether it reduces readmissions—requires prospective evaluation.